

Vaibhav Gollapalli

469-794-9743 | vaibhavgollapalli5@gmail.com | linkedin.com/in/vaibhavgollapalli | vaibhav100968.github.io | Github

EDUCATION

Texas A&M University

Bachelor of Science in Computer Science

College Station, TX

Expected Graduation: May 2028

- **Relevant Coursework:** Data Structures & Algorithms, Discrete Mathematics, Linear Algebra, Multivariable Calculus, Differential Equations, Statistics

EXPERIENCE

Software Engineering Intern

May 2026 – Present

Soularis

Austin, TX

- Built a **pgvector** search service (HNSW-indexed) with sub-second semantic queries, lifting retrieval relevance **70%+**.
- Engineered a **RAG** service with schema validation, holding **100%** response-schema compliance at **2.3s** p95 latency.
- Consolidated **25** backend endpoints under one Railway schema, cutting parsing logic **20%** and error rate **15%**.

Research Fellow

Nov. 2024 – Present

UNT Mixed Realities Lab

Denton, TX

- Built a real-time streaming service over **MQTT + WebSockets**, delivering sensor data to clients at **512ms** latency.
- Engineered a **Python** pipeline processing **16,904** records/run with automated ingestion, transformation, and reporting.
- Developed a **REST API** with **8+** endpoints serving datasets to visualization and model clients through one layer.

AI Systems Engineering Intern

May 2025 – Aug. 2025

CORE (YC S23)

Remote

- Optimized backend APIs and orchestration, raising workflow reliability **30%** across distributed multi-service systems.
- Built a distributed **Model Context Protocol (MCP)** automating workflows across **4+** integrated backend services.
- Scaled backend infrastructure and internal debugging tools, cutting repetitive manual engineering work by **30%**.

PROJECTS

Mean Eats | Swift, SwiftUI, Supabase, AWS Lambda, BeautifulSoup4

Nov. 2025 – Present

- Shipped an App Store dining app to **300+** users across **5** halls with live menus and meal-plan tracking.
- Automated **5** dining scrapers on serverless **AWS Lambda** + EventBridge, syncing menus with **180+** days uptime.
- Built an offline cache and on-device recommendations, cutting redundant API calls **60%+** on every launch.

Glyph | Swift, SwiftUI, Core ML, Fireworks AI, PyTorch

Jan. 2026 – Present

- Built an iPad tutor app with a real-time **Apple Pencil** canvas, on-device **Apple Vision** OCR, and **6** study widgets.
- Engineered an on-device **Core ML** inference pipeline generating handwriting in real time with no server round-trips.
- Cut LLM inference cost **40%** with an on-device/cloud routing cascade, running arithmetic locally at **\$0**.

OpenGreyWolfOptimizer | Python, NumPy, mpi4py, pytest

Jul. 2025 – Present

- Shipped an open-source **GWO** package with an sklearn-style **.fit()** API, full type hints, and a JOSS paper draft.
- Built a benchmark harness across **4** standard test functions, validating convergence in under **0.03s** per run.
- Added optional **MPI** parallel fitness evaluation plus **13** pytest cases and CI across **3** Python versions.

TECHNICAL SKILLS

Languages: Java, Python, C++, C#, TypeScript, JavaScript, Swift, SQL, HTML, CSS

Frameworks: React, React Native, Node.js, Express, FastAPI, SwiftUI

Tools & Platforms: Git, Docker, AWS, GitHub Actions, JIRA, Microsoft Excel, Postman, PostgreSQL

Practices & Testing: CI/CD, Agile, Unit Testing, REST APIs, Pytest, Playwright

AI & ML: PyTorch, TensorFlow, Core ML, scikit-learn, RAG, LangChain, prompt engineering

Leadership:

- **HackTAMS** (Director, 2024–2026): Directed **2** years, scaling **30** to **230+** attendees and raising **\$20K+**, top Texas hackathon.
- **Microsoft Copilot Ambassador** (2026): **1 of 150** nationwide, leading campus AI-tool adoption, workshops, and events.

Publications:

- “Federated TinyML for IoMT Anomaly Detection via MQTT Edge,” **First author**, *IEEE iSES 2025* ([IEEE](#)).
- “Hybrid Windkessel-Neural Noninvasive Blood Pressure Monitoring,” **First author**, *Preprint* ([arXiv](#)).
- “Cognitive Energy Modeling with EEG and WGAN-GP,” **First author**, *IEEE SMC 2026* ([arXiv](#)).
- “StrokeSentry: Edge AI Mobile App for Rapid Stroke Detection,” Second author, *IEEE ICBAIE 2025* ([IEEE](#)).